

Comparative Safety of Long-Acting vs. Short-Acting Erythropoiesis-Stimulating Agents Among Patients Undergoing Hemodialysis

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Both short-acting (epoetin alfa or beta) and long-acting (darbepoetin alfa or PEG-epoetin) erythropoiesis-stimulating agents (ESAs) are commonly prescribed for patients with kidney failure undergoing maintenance hemodialysis. We compared the risks of major adverse cardiovascular events (MACE) and of all-cause mortality associated with receipt of short- vs. long-acting ESAs. This retrospective cohort analysis included Medicare hemodialysis beneficiaries aged ≥ 18 years in the United States Renal Data System from January 2015 to December 2017. We included adults who survived > 90 days after initiating hemodialysis and received either short- or long-acting ESAs. Outcomes were MACE (first occurrence of stroke, acute myocardial infarction, or cardiovascular-related mortality) and all-cause mortality. After stabilized inverse probability of treatment weighting, Cox proportional hazards regression models were used to estimate the hazard ratio (HR) and 95% confidence interval (CI) for each outcome. Of 68,607 patients (mean age: 65 years, 45% females), 33,658 (49%) received long-acting ESAs and 34,949 (51%) received short-acting ESAs. There was no difference in the risk of MACE associated with receipt of short- vs. long-acting ESAs (HR: 1.02 (95% CI: 0.98–1.08)). However, long-acting (vs. short-acting) ESA receipt was associated with a lower risk of all-cause mortality (HR: 0.91 (95% CI: 0.87–0.96)). Compared with short-acting ESAs, long-acting ESAs were associated with a lower risk of all-cause mortality, with no difference in the risk of MACE. Future studies with a longer follow-up are needed to confirm these findings.

Study Highlights

WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

☑ Patients with kidney failure undergoing maintenance hemodialysis receive short-acting (epoetin alfa or beta) and long-acting (darbepoetin alfa or PEG-epoetin) erythropoiesis-stimulating agents (ESAs). However, their effects on major adverse cardiovascular events (MACE) and all-cause mortality risks vary.

WHAT QUESTION DID THIS STUDY ADDRESS?

☑ This study aimed to assess the risk of MACE, including stroke, acute myocardial infarction (AMI), and cardiovascular-related mortality, and all-cause mortality among hemodialysis patients receiving short- vs. long-acting ESAs.

WHAT DOES THIS STUDY ADD TO OUR KNOWLEDGE?

☑ This nationwide registry cohort study of 68,607 patients receiving hemodialysis found no differences in the risk of

MACE associated with receipt of short- vs. long-acting ESAs (HR: 1.02 (95% CI: 0.98–1.08)). However, long-acting (vs. short-acting) ESA receipt was associated with a lower risk of all-cause mortality (HR: 0.91 (95% CI: 0.87–0.96)), particularly for PEG-epoetin recipients (HR: 0.84 (95% CI: 0.78, 0.90)).

HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY OR TRANSLATIONAL SCIENCE?

☑ There may be mortality benefits associated with the use of PEG-epoetin compared with the use of short-acting ESAs for patients undergoing hemodialysis.

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Anemia is a common and prognostically significant complication affecting patients with kidney failure requiring hemodialysis.¹ Erythropoiesis-stimulating agents (ESAs), including those that are short-acting (epoetin alfa and beta) and long-acting (darbepoetin alfa and methoxy polyethylene glycol-epoetin beta (PEG-epoetin)) are used interchangeably for the treatment of anemia in patients with kidney failure.² The use of ESAs has improved the quality of life for patients with kidney failure and decreased their need for blood transfusions.¹ Because long-acting ESAs have a longer half-life than short-acting ESAs, the former is administered less frequently and may contribute to more stable hemoglobin levels.^{3–5} Short-acting epoetin is administered twice weekly and long-acting darbepoetin alfa is administered once a week, whereas long-acting PEG-epoetin is conventionally administered once every 2 or 4 weeks.⁶

Although short- and long-acting ESAs are commonly used to maintain hemoglobin levels for the treatment of anemia in patients with kidney failure, data regarding the comparative safety of these agents in the United States (US) are limited. Using data from the US Renal Data System (USRDS), Winkelmayer and colleagues evaluated the risk of major adverse cardiovascular events (MACEs) and all-cause mortality among patients with kidney failure who received darbepoetin alfa (long-acting ESA) vs. those who received epoetin alfa (short-acting ESA).⁷ The comparison groups in that study were hemodialysis facilities that predominantly utilized darbepoetin alfa matched with corresponding facilities that predominately utilized epoetin alfa. The investigators concluded that the risks of MACE (adjusted hazard ratio (aHR): 1.10 (95% confidence interval (CI): 0.96, 1.25)) and all-cause mortality (aHR: 1.05 (95% CI: 0.99, 1.12)) were similar among recipients of darbepoetin alfa and epoetin alfa. In addition, a meta-analysis of randomized clinical trials found no difference in the risk of mortality for patients receiving epoetin vs. darbepoetin. However, that meta-analysis was limited by a small sample size (2000 patients in nine trials), a short follow-up duration (<1 year), and inadequate power.⁸ Another meta-analysis also limited by a small sample size similarly did not find any difference in the risk of mortality between epoetin and darbepoetin use.⁹ Notably, neither the study by Winkelmayer et al. nor the meta-analyses assessed risks among patients receiving PEG-epoetin, another long-acting ESA initially approved by the US Food and Drug Administration in 2007 but not marketed in the United States until 2014.¹⁰ Since 2015, the use of PEG-epoetin has increased substantially and has exceeded the use of darbepoetin among dialysis patients.¹¹ Given the pharmacokinetic differences and frequencies of administration among the various types of ESAs, their prognostic effects may vary. Thus, the present study aimed to assess the risk of MACE, including stroke, acute myocardial infarction (AMI) and cardiovascular-related mortality, and all-cause mortality among hemodialysis patients receiving short- vs. long-acting ESAs.

METHODS

Study design and data source

We conducted a cohort study using data from the USRDS between January 1, 2015, and December 31, 2017, for Medicare-eligible

patients receiving hemodialysis. The USRDS is a national registry of patients with kidney failure based on Medicare claims submitted to the Centers for Medicare and Medicaid Services by providers that include Medicare enrollment history, Medicare Parts A and B claims, and death information.¹² This database has near-universal inclusion of patients with kidney failure in the United States. Institutional review board approval for this study was obtained from the University of Florida.

Study population

We included patients aged ≥ 18 years who initiated hemodialysis between January 1, 2015, and December 31, 2017. Patients were required to survive at least 90 days after the initiation of hemodialysis (inclusive of the mandated 90-day waiting period for Medicare benefits eligibility by the Centers for Medicare and Medicaid Services) and to be continuously enrolled in Medicare Parts A and B through Day 91. Patients were also required to use short-acting ESAs (identified by Healthcare Common Procedure Coding System (HCPCS) codes J0885, J0886, Q4081, Q9972, and Q9973 and by Revenue Codes 0634, 0635, and 0636) or long-acting ESAs (identified by HCPCS codes J0881, J0882, and J0887 and by Revenue Codes 0634, 0635, and 0636) within 6 months after Day 91 of hemodialysis. Based on their first claim for ESAs after Day 91 of hemodialysis, patients were placed in one of two groups: short-acting ESA (epoetin alfa) or long-acting ESA (PEG-epoetin and darbepoetin) recipients. Epoetin beta is not approved in the United States and, thus, was not part of this study. The first service date of ESAs was defined as the index date of cohort entry. Patients were considered to be continuously exposed as long as they continued to receive short- or long-acting ESAs without a gap of ≥ 45 days. This 45-day grace period was selected to account for the carryover effects of discontinued ESA medications and to allow sufficient time for an outcome to occur. Patients were allowed to switch within the ESA categories (i.e., from PEG-epoetin darbepoetin, and vice versa), but not across ESA categories. Patients were excluded if they underwent kidney transplantation at any time before the index date, had missing values in the Medical Evidence Report for the presence of comorbidities on the index hemodialysis date, or had a MACE occurrence in the 90-day waiting period.

Study outcome and follow-up

The primary outcome was MACE, defined as the first occurrence of stroke (identified by *International Classification of Diseases, Ninth Revision, Clinical Modification* (ICD-9-CM) codes 430, 431, 433.x1, 434.x1, and 436 or ICD-10-CM codes I60, I61, I63, and I64), or AMI (identified by ICD-9-CM code 410.x or ICD-10-CM code I21.x) in the primary position of hospital discharge claims, or cardiovascular-related mortality, defined as death due to MI, ischemic stroke, intracranial hemorrhage, sudden cardiac death, or heart failure. Those identifications of stroke and AMI have been validated previously with positive predictive values $> 90\%$.^{13,14} The secondary outcomes were individual components of the primary outcome, namely, stroke, AMI, cardiovascular-related mortality, and all-cause mortality.

Follow-up began on the index date and continued until the first occurrence of an outcome, discontinuation of therapy (including a 45-day grace period), a switch to the comparator (i.e., from short- to long-acting ESA or vice versa), end of Medicare Parts A and B enrollment, or end of the study period (December 31, 2017).

Covariates

We ascertained patient demographic characteristics (e.g., age, sex, US geographic region, and race and ethnicity) from patient files. We collected serum albumin level, body mass index, hemoglobin level, tobacco use, primary reason for dialysis, and medical comorbidities, including atherosclerotic heart disease, hypertension, heart failure, cerebrovascular

disease, peripheral artery disease, cancer, chronic obstructive pulmonary disease, and diabetes mellitus, from the Medical Evidence Report at the time of dialysis initiation.

Sensitivity and subgroup analyses

We conducted several sensitivity and subgroup analyses to assess the robustness of our study findings. First, we modified the grace period from 45 days to 60 days. Second, we replicated the main analysis but compared epoetin alfa vs. darbepoetin and separately compared epoetin alfa vs. PEG-epoetin. For that analysis, in addition to the aforementioned censoring criteria, we also censored patients when they switched to the other long-acting ESA (i.e., from darbepoetin to PEG-epoetin and vice versa). Additionally, we conducted several prespecified subgroup analyses to assess heterogeneity of treatment effect by stratifying the following baseline covariates: age (< 65 vs. ≥ 65 years), sex (male vs. female), race and ethnicity (White vs. Black vs. other) and history of ESA use (yes vs. no).

Statistical analysis

Descriptive statistics of patient baseline characteristics were compared between groups with short- vs. long-acting ESAs by assessing means for continuous variables and proportions for categorical variables. To adjust for differences in patient baseline characteristics and clinical risk factors between the two groups, we used stabilized inverse probability of treatment weighting (sIPTW) based on propensity scores. We assessed the effectiveness of sIPTW by cross-tabulating baseline characteristics before and after sIPTW by the two groups (i.e., short-acting vs. long-acting ESAs) and reported standardized mean differences. Variables with absolute standardized mean differences < 0.1 were considered well-balanced. For the sensitivity analysis comparing darbepoetin vs. short-acting ESAs and PEG-epoetin vs. short-acting ESAs, sIPTW was also used to adjust for differences in patient baseline characteristics and clinical risk factors between the two comparison groups. Cox proportional hazards regression models with sIPTW were used to estimate HRs and 95% CIs to compare the risk of study outcomes between short- and long-acting ESA recipients. The proportional hazard assumption was tested using Schoenfeld residuals. We used SAS 9.4 (SAS Institute, Cary, NC, USA) for data management and analyses. A *P* value < 0.05 was considered to be statistically significant.

RESULTS

Cohort characteristics

After applying inclusion and exclusion criteria, we identified 68,607 patients who received either long-acting or short-acting ESAs. Of them, 33,658 patients (49%) received long-acting ESAs and 34,949 patients (51%) received short-acting ESAs. After sIPTW, all covariates were well balanced with absolute standardized mean differences < 0.1, and the cohort included 45% women and had a mean age of 65 years (Table 1).

Risk for MACE and its components

We identified 2,477 episodes of MACE during a mean follow-up of 237 days in the short-acting ESA group and 3,653 episodes of MACE during a mean follow-up of 343 days in the long-acting ESA group. The crude incidence rates of MACE were 109.2 and 115.4 per 1,000 person-years in the short-acting and long-acting ESA groups, respectively. In the Cox proportional hazards regression model after sIPTW, there was no difference in the risk of MACE associated with receipt of short- vs. long-acting ESAs (HR: 0.99 (95% CI: 0.93, 1.06)) (Table 2).

No differences in the risks of the individual components of MACE were observed between the two groups, including stroke (HR: 0.98 (95% CI: 0.85, 1.14)), AMI (HR: 0.90 (95% CI: 0.81, 1.01)), and cardiovascular-related mortality (HR: 1.05 (95% CI: 0.97, 1.14)).

Risk for all-cause mortality

We identified 3,882 all-cause mortality events during a mean follow-up of 243 days in the short-acting ESA group, and 2,527 events during a mean follow-up of 170 days in the long-acting ESA group. The crude incidence rates of all-cause mortality were 166.5 and 161.4 per 1,000 person-years in the short-acting and long-acting ESA groups, respectively. In the Cox proportional hazards regression model after sIPTW, receipt of long-acting ESAs was associated with significantly lower risk of all-cause mortality (HR: 0.91 (95% CI: 0.87, 0.96)) compared with receipt of short-acting ESAs (Table 2).

Sensitivity and subgroup analysis

Findings from sensitivity analyses remained consistent with those from the main analyses when we changed the grace period from 45 days to 60 days (Table 3). There were no differences in the risks of MACE (HR: 1.06 (95% CI: 0.98, 1.15)) or all-cause mortality (HR: 1.00 (95% CI: 0.93, 1.06)) associated with receipt of darbepoetin vs. short-acting ESAs. Although no difference in the risk of MACE was observed for receipt of PEG-epoetin (HR: 0.94 (95% CI: 0.87, 1.02)), its receipt was associated with significantly lower risk of all-cause mortality (HR: 0.84 (95% CI: 0.78, 0.90)) compared with receipt of short-acting ESAs (Table 3). In subgroup analyses, we found no evidence of treatment effect heterogeneity among patients ≤ 65 years vs. > 65 years of age, across race and ethnicity groups, nor in patients with vs. without a history of ESA use, for either outcome (p-interaction > 0.05 for all; Figure 1).

DISCUSSION

In this population-based retrospective cohort study, we compared the risk of MACE and of all-cause mortality among patients with kidney failure who received short-acting ESAs vs. those who received long-acting ESAs. Our findings suggested no difference in the risk of MACE. However, patients receiving long-acting ESAs, particularly PEG-epoetin, had lower risk of all-cause mortality compared with those receiving short-acting ESAs. These findings remained consistent across subgroup analyses by sex (female vs. male), race and ethnicity (Black vs. White vs. other), age group (< 65 vs. ≥ 65 years), or history of ESA use (yes vs. no).

ESAs are the cornerstone for the treatment of anemia in patients with chronic kidney disease. Both short- and long-acting ESAs reduce the need for blood transfusion, reduce anemia-related symptoms, and improve the quality of life.^{15–17} Although ESAs have been available for decades, data regarding the long-term clinical outcomes among patients with kidney failure who receive long- vs. short-acting ESAs are limited. Our findings were consistent with those from a study conducted by Winkelmayer and colleagues.⁷ In that study, the investigators conducted a quasi-experimental

Table 1 Baseline characteristics of the study cohort

Characteristic	Before sIPTW		SMD	After sIPTW		SMD
	Long-acting ESA	Short-acting ESA		Long-acting ESA	Short-acting ESA	
No. of total unique patients	68,607			67,606		
No. of unique patients	33,658	34,949		33,653	33,953	
Female sex, <i>n</i> (%)	15,244 (45.3)	15,578 (44.6)	0.01	15,115 (44.9)	30,811 (44.9)	0.00
Age, years, mean (SD)	66.2±14.1	65.2±14.7	−0.07	65.7±14.3	65.7±14.5	0.00
Race, <i>n</i> (%)			0.07			0.00
White	23,350 (69.4)	22,933 (65.6)		22,701 (67.5)	23,571 (67.4)	
Black	8,404 (25.0)	9,570 (27.4)		8,791 (26.1)	9,141 (26.2)	
Other/Unknown/missing	1,904 (5.7)	2,446 (7.0)		2,161 (6.4)	2,241 (6.4)	
US region, <i>n</i> (%)			0.18			0.00
Northeast	7,077 (21.0)	6,462 (18.5)		6,657 (19.8)	6,919 (19.8)	
Midwest	6,317 (18.8)	6,345 (18.2)		6,198 (18.4)	6,439 (18.4)	
West	4,927 (14.6)	7,612 (21.8)		6,149 (18.3)	6,388 (18.3)	
South	14,806 (44.0)	14,213 (40.7)		14,235 (42.3)	14,780 (42.3)	
Other/unknown/missing	531 (1.6)	317 (0.9)		414 (1.2)	427 (1.2)	
Primary cause of kidney failure, <i>n</i> (%)			0.04			0
Diabetes	17,639 (52.4)	17,585 (50.3)		17,278 (51.3)	17,943 (51.3)	
Hypertension	10,038 (29.8)	10,981 (31.4)		10,297 (30.6)	10,700 (30.6)	
Glomerulonephritis	2,112 (6.3)	2,126 (6.1)		2,091 (6.2)	2,169 (6.2)	
Cystic kidney	470 (1.4)	447 (1.3)		452 (1.3)	468 (1.3)	
Other	3,399 (10.1)	3,810 (10.9)		3,535 (10.5)	3,672 (10.5)	
Mean BMI, kg/m ² , mean±SD	29.8±8.0	29.6±8.0	−0.03	29.7±8.0	29.7±8.1	0.00
Mean hemoglobin, g/dL, mean±SD	9.2±1.7	9.1±1.8	−0.01	9.1±1.7	9.1±1.8	0.00
Albumin, g/dL, mean±SD	3.1±1.0	3.1±1.3	−0.01	3.1±1.0	3.1±1.7	0.00
Tobacco use, <i>n</i> (%)	2,033 (6.0)	2,248 (6.4)	0.02	2,103 (6.3)	2,182 (6.2)	0.00
Comorbidity, <i>n</i> (%)						
Atherosclerotic heart disease	5,754 (17.1)	4,872 (13.9)	−0.09	5,220 (15.5)	5,419 (15.5)	0.00
Heart failure	10,980 (32.6)	10,737 (30.7)	−0.04	10,650 (31.6)	11,060 (31.6)	0.00
Cerebrovascular disease	3,339 (9.9)	3,248 (9.3)	−0.02	3,245 (9.6)	3,370 (9.6)	0.00
Peripheral vascular disease	4,079 (12.1)	3,732 (10.7)	−0.05	3,834 (11.4)	3,980 (11.4)	0.00
Chronic obstructive pulmonary disease	3,606 (10.7)	3,382 (9.7)	−0.03	3,429 (10.2)	3,560 (10.2)	0.00
Diabetes mellitus	20,902 (62.1)	21,490 (61.5)	−0.01	20,788 (61.8)	21,591 (61.8)	0.00
Hypertension	30,243 (89.9)	31,223 (89.3)	−0.02	30,148 (89.6)	31,309 (89.6)	0.00
Cancer	2,372 (7.0)	2,628 (7.5)	−0.02	2,456 (7.3)	2,550 (7.3)	0.00

BMI, body mass index; SMD, standardized mean difference; SD, standard deviation; sIPTW, stabilized inverse probability of treatment weighting.

cohort study to compare the risk of MACE among users of epoetin alfa and darbepoetin alfa in patients with kidney failure who were undergoing hemodialysis in the United States.⁷ Findings from that study suggested that the risk of all-cause mortality and cardiovascular mortality was also compared between users of epoetin alfa and darbepoetin alfa. That study found no significant differences in the risk of all-cause mortality (HR: 1.06; 95% CI: 1.00–1.13), cardiovascular mortality (HR: 1.05; 95% CI: 0.94–1.16) or MACE (HR: 1.10; 95% CI: 0.96–1.25). Similar to that study, our study showed no difference in the risk of MACE associated with receipt of short-acting epoetin alfa vs. long-acting darbepoetin alfa.

However, our present study extended those findings by including PEG-epoetin in the group of patients receiving long-acting ESAs. In contrast to the results of the study by Winkelmayer and colleagues,⁷ our findings indicated that receipt of long-acting ESAs was associated with lower risk of mortality compared with receipt of short-acting ESAs.

The mechanisms underlying different mortality risks among patients receiving ESAs are unclear but may be related to hemoglobin level variations and their association with increased adverse outcomes. The inability to maintain hemoglobin levels within target ranges and high fluctuations in these levels are associated

Table 2 Risk for major adverse cardiovascular events (MACE) and all-cause mortality among patients receiving short-acting and long-acting erythropoiesis-stimulating agents (ESAs)

Outcome	Short-acting ESAs (N = 34,949)			Long-acting ESAs (N = 33,658)			Long-acting ESA vs. Short-acting ESA (reference) Adjusted hazard ratio ^a (95% CI)
	Number of events	Person-time (years)	Incident rate per 1,000 person-years	Number of events	Person-time (years)	Incident rate per 1,000 person-years	
MACE	2,477	22,690	109.2	1,786	15,374	116.2	0.99 (0.93, 1.06)
Stroke	462	23,495	19.7	324	15,851	20.4	0.98 (0.85, 1.14)
Acute myocardial infarction	831	23,252	35.7	543	15,768	34.4	0.90 (0.81, 1.01)
Cardiovascular mortality	1,421	23,313	61.0	1,070	15,656	68.3	1.05 (0.97, 1.14)
All-cause mortality	3,882	23,313	166.5	2,527	15,656	161.4	0.91 (0.87, 0.96)

^aAdjusted for age, sex, race, region, serum albumin levels, body mass index (BMI), hemoglobin (g/dL), tobacco use, primary reason for dialysis, medical comorbidities including atherosclerotic heart disease, hypertension, heart failure, cerebrovascular disease, peripheral artery disease (PVD), cancer, chronic obstructive pulmonary disease (COPD), and diabetes mellitus (DM), and history of ESA.

Table 3 Sensitivity analysis: Risk for major adverse cardiovascular events (MACE) and all-cause mortality

Outcome	Short-acting ESAs (N = 34,949)			Long-acting ESAs (N = 33,658)			Long-acting ESA vs. Short-acting ESA (reference) Adjusted hazard ratio ^a (95% CI)
	Number of events	Person-time (years)	Incident rate per 1,000 person-years	Number of events	Person-time (years)	Incident rate per 1,000 person-years	
Sensitivity analysis: 60-day grace period							
MACE	2,749	24,668	111.4	2,187	18,289	119.6	1.01 (0.95, 1.07)
Stroke	515	25,956	19.8	396	19,245	20.6	0.98 (0.86, 1.12)
Acute myocardial infarction	894	25,683	34.8	652	19,118	34.1	0.92 (0.83, 1.02)
Cardiovascular mortality	1,614	25,386	63.6	1,336	18,674	71.5	1.06 (0.98, 1.14)
All-cause mortality	4,555	25,386	179.4	3,304	18,674	176.9	0.93 (0.89, 0.97)
Sensitivity analysis: darbepoetin vs. short-acting ESAs							
MACE	2,490	22,720	109.6	894	7,295	122.5	1.06 (0.98, 1.15)
All-cause mortality	3,902	23,343	167.2	1,291	7,444	173.4	1.00 (0.93, 1.06)
Sensitivity analysis: PEG-epoetin vs. short-acting ESAs							
MACE	2,483	22,704	109.4	838	7,524	111.4	0.94 (0.87, 1.02)
All-cause mortality	3,894	23,327	166.9	1,155	7,633	151.3	0.84 (0.78, 0.90)

^aAdjusted for age, sex, race, region, serum albumin levels, body mass index (BMI), hemoglobin (g/dL), tobacco use, primary reason for dialysis, medical comorbidities including atherosclerotic heart disease, hypertension, heart failure, cerebrovascular disease, peripheral artery disease (PVD), cancer, chronic obstructive pulmonary disease (COPD), and diabetes mellitus (DM) and history of ESA.

with increased risk of cerebrovascular disease, cardiovascular disease, infection, and hospitalization.¹⁸ With their longer half-life, long-acting ESAs may have less hemoglobin variability,⁴ enabling them to maintain the stable hemoglobin level range of 11–12 g/dL that is recommended by the National Kidney Foundation Kidney Disease Outcome Quality Initiative (KDOQI).¹⁹ Lee et al.²⁰ found 10% increased risk of all-cause mortality when hemoglobin levels decreased below the target range. Another study reported 5% reduced mortality risk for every 1 g/dL increase in hemoglobin toward the target range.²¹ They also found that patients with hemoglobin concentrations lower than 10 g/dL had 29% higher risk of hospitalization than those in the 11–12 g/dL range. However, higher target hemoglobin levels related to ESA therapy are associated with certain risks, such as stroke and thrombosis.^{9,22,23}

Hence, the recommendation by KDOQI to avoid hemoglobin targets above 13 g/dL.¹⁹ The findings of these studies suggest the need for risk–benefit balance in the use of ESAs. With short-acting ESAs having a substantially shorter half-life than long-acting ESAs, short-acting ESAs may be less likely to achieve target hemoglobin ranges, leading to more cardiovascular complications.⁴

Our findings appear to be inconsistent with the results from an observational study using data from the Japanese Society for Dialysis Therapy Renal Data Registry that evaluated the risk of all-cause mortality among individuals receiving short-acting or long-acting ESAs.¹⁵ During the 2-year follow-up in that study, the authors concluded that compared with receipt of short-acting ESA, receipt of long-acting ESAs was associated with 13% higher risk of mortality (aHR: 1.13 (95% CI: 1.09, 1.16)). This inconsistency

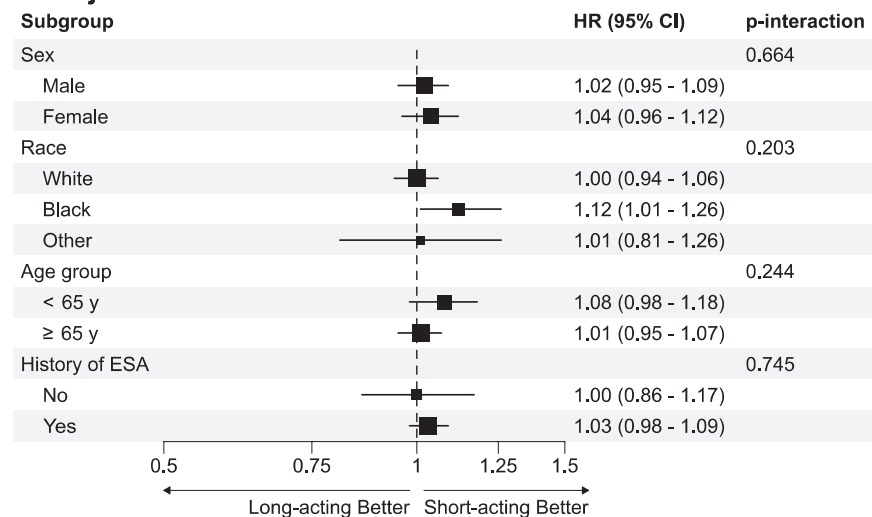
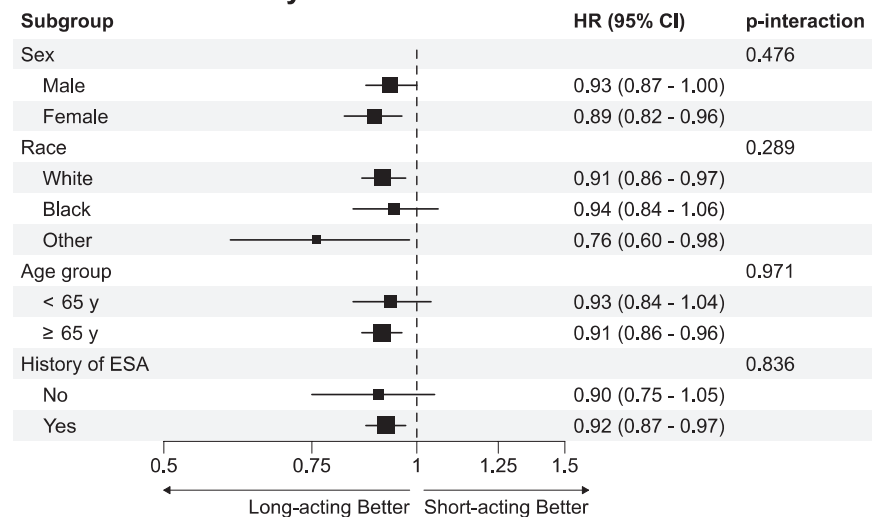
(a) Major Adverse Cardiovascular Events**(b) All-cause Mortality**

Figure 1 Results of subgroup analyses comparing the risks of major adverse cardiovascular events and all-cause mortality among patients receiving short-acting vs. long-acting erythropoiesis-stimulating agents (ESAs).

between their findings and our results may be due to the differences in the ESAs that were studied. That study included epoetin κ (short-acting ESA) and an erythropoietin receptor activator (long-acting ESA), neither of which are available in the United States and thus were not included in the present study. Another potential reason for the difference in findings may be the differences in the maximum allowable dose across the different types of ESAs in Japan.¹¹ For instance, Japan has a unique anemia treatment guideline compared with other countries: the maximum dose prescribed for short-acting ESAs is capped (i.e., 9,000 units/week), but there is no dose restriction for long-acting ESAs. It is plausible that patients treated with a long-acting ESA in Japan received a greater amount on average of an equivalent ESA than a short-acting ESA and may have been hyporesponsive to the ESA.³ This scenario may explain the increased risk of mortality associated with long-acting ESAs compared with short-acting ESAs observed in the study

conducted in Japan because higher doses of ESAs are known to be associated with more adverse events. The findings of our study are also inconsistent with those in a recently published study that compared mortality rates among patients prescribed long- vs. short-acting ESAs using data from the Dialysis Outcomes and Practice Patterns Study in three geographical regions: North America, Europe, and Japan.³ For a median follow-up of 14 months, the authors reported that there was no difference in mortality rates associated with receipt of long- vs. short-acting ESAs North America. These findings may be explained by the difference in the rates of PEG-epoetin use (i.e., 10% in the previous study vs. 72% in our study). We conducted a sensitivity analysis excluding patients with PEG-epoetin use to assess the robustness of our findings. Similar to the main analysis, there were no significant differences in the risks of MACE or all-cause mortality between short-acting ESAs and darbepoetin. However, a significantly decreased risk of all-cause

mortality was observed in the PEG-epoetin group compared with the short-acting ESA group, suggesting that PEG-epoetin contributed to the lower risk of all-cause mortality in the long-acting ESA group. This decreased risk of all-cause mortality associated with long-acting ESA use may be explained by the stable utilization of iron for erythropoiesis and, in turn, reduced induction of hepcidin. Hepcidin is a peptide hormone that regulates iron homeostasis, and the levels of hepcidin increase during inflammation, which is common in hemodialysis patients.^{24,25} Suppression of hepcidin leads to mobilization of iron from macrophage storage, resulting in stable target hemoglobin levels.^{25,26} Further studies on mortality risks and benefits related to PEG-epoetin use are needed to confirm our findings.

The strengths of the present study included the use of data from the USRDS, a national registry that includes most patients with kidney failure in the United States. The use of this registry provided a large sample size to generate high study power, enabling the detection of significant effects. Additionally, this study incorporated PEG-epoetin into the assessments of long-acting ESA safety and reported findings that suggested a lower risk of all-cause mortality compared with short-acting ESAs. However, our study had several limitations. First, we relied on *ICD-9-CM* and *ICD-10-CM* codes to identify MACE outcomes. Although we used previously validated codes,^{13,14} it is possible that incomplete, missing, or miscoded claims may have impacted the study findings. Nevertheless, coding errors were likely to have been similarly distributed between the two groups. Second, we restricted our cohort to patients who survived at least 90 days after their first hemodialysis treatment. The lack of a new-user design may have led to under-ascertainment of outcomes that occurred early in therapy.²⁷ Considering the known high mortality rate in the first 90 days of dialysis, it is important to highlight that our results are generalizable only to patients who survive the first 90 days. Third, we did not assess hemoglobin levels at cohort entry, which would affect the dose of ESA needed to reach the optimal range for hemoglobin levels. Finally, because this was an observational study, the results cannot be used to draw causal inferences.

In conclusion, this population-based cohort study of patients with kidney failure detected no difference in the risk of MACE associated with receipt of short- vs. long-acting ESAs. However, there may be mortality benefits associated with the use of PEG-epoetin compared with short-acting ESAs for patients undergoing hemodialysis. Future studies with longer follow-up are needed to confirm these findings.

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CONFLICT OF INTEREST

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AUTHOR CONTRIBUTIONS

R.D., H.P., S.M.S., and I.U. wrote the manuscript. R.D., S.M.S., and H.P. designed the research. R.D., H.P., M.R., A.M.S., R.M., N.J., and I.U. performed the research. R.D. and H.P. analyzed the data.

DISCLAIMER

The data reported here have been supplied by the United States Renal Data System (USRDS). The interpretation and reporting of these data are the responsibility of the author(s) and in no way should be seen as an official policy or interpretation of the US government.

DATA AVAILABILITY STATEMENT

Data are available from the United States Renal Data System (USRDS) under data use agreement provisions. As per the data use agreement, the relevant limited data sets cannot be made publicly available.

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